DreamBooth: Fine-Tuning Text-to-Image Diffusion Models for Subject-Driven Generation

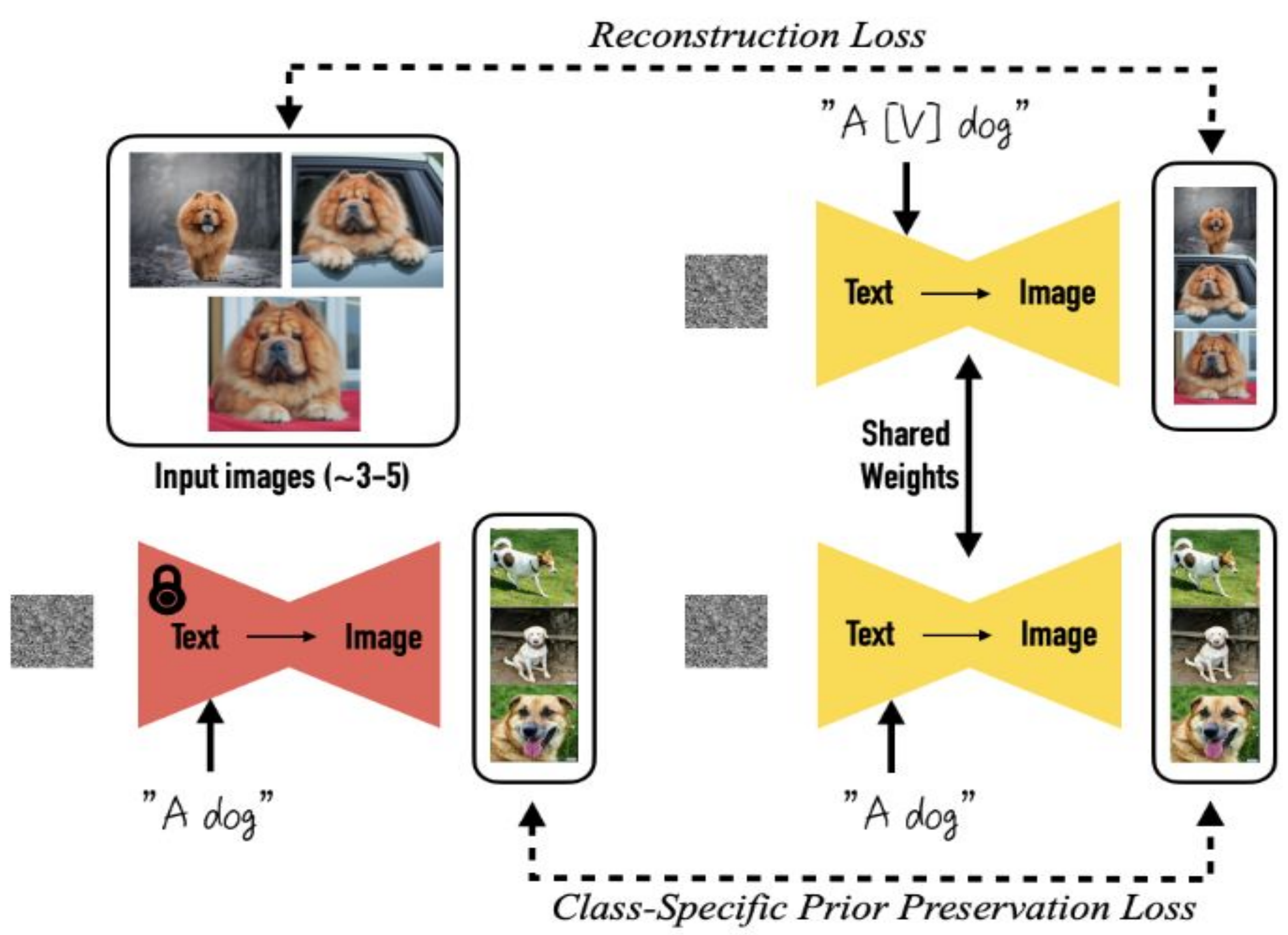
Jaxin Lu, Haotian Pu, Julia Lee

Background

Motivation

- Text-to-image models (DALL-E 2, Imagen) can't reproduce specific subjects.
- DreamBooth personalizes models from just 3–5 photos and generates image with same identity in new contexts (poses, scenes, styles).

Model Structure



Our Goal

1. Replicate subject & prompt fidelity
2. Reproduce prior preservation loss & class-prior ablations
3. Generate qualitative results
4. Extend: evaluate on new subjects

References

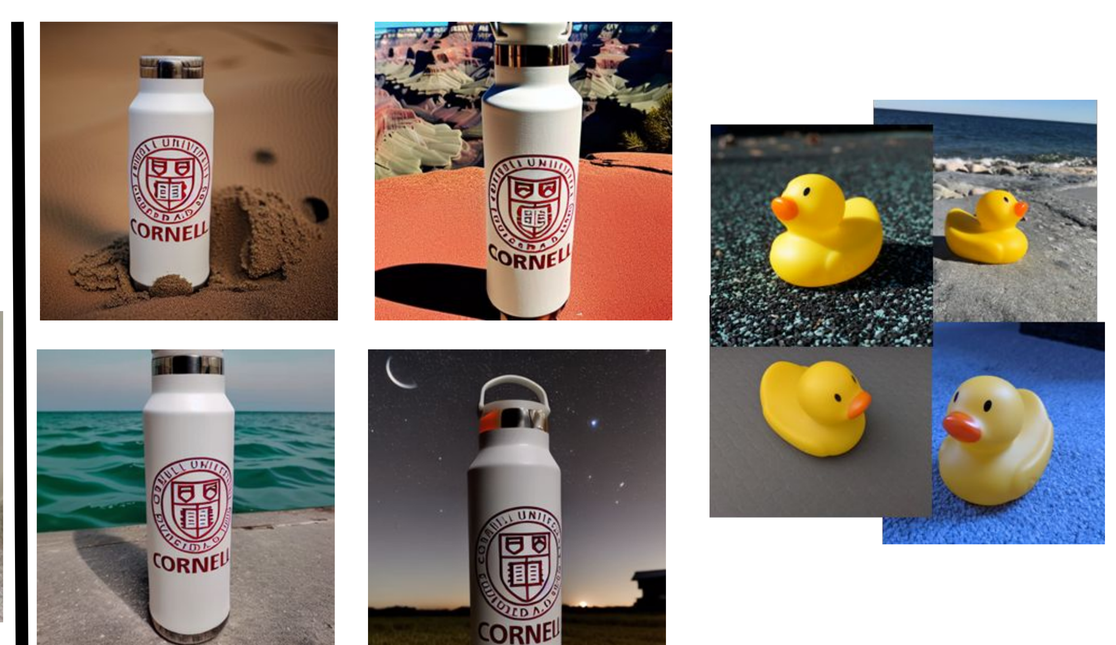
Ruiz, N., et al. (2023) "DreamBooth: Fine Tuning Text-to-Image Diffusion Models for Subject-Driven Generation"

Experiment Settings

Model	stable-diffusion-v1-5
Dataset	DreamBooth Dataset (GitHub, ~100 MB)
Hardware	Google Colab Pro (A100 GPU)

Results

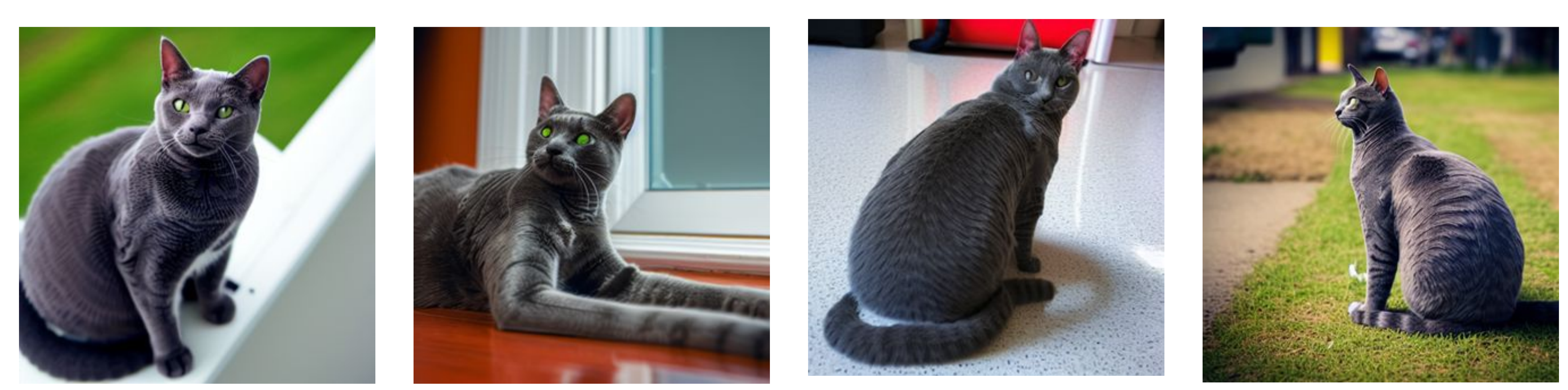
Input Images



Background changes

Outfit changes

Input Images



Top view **Bottom view** **Back view** **Side view**

Metrics

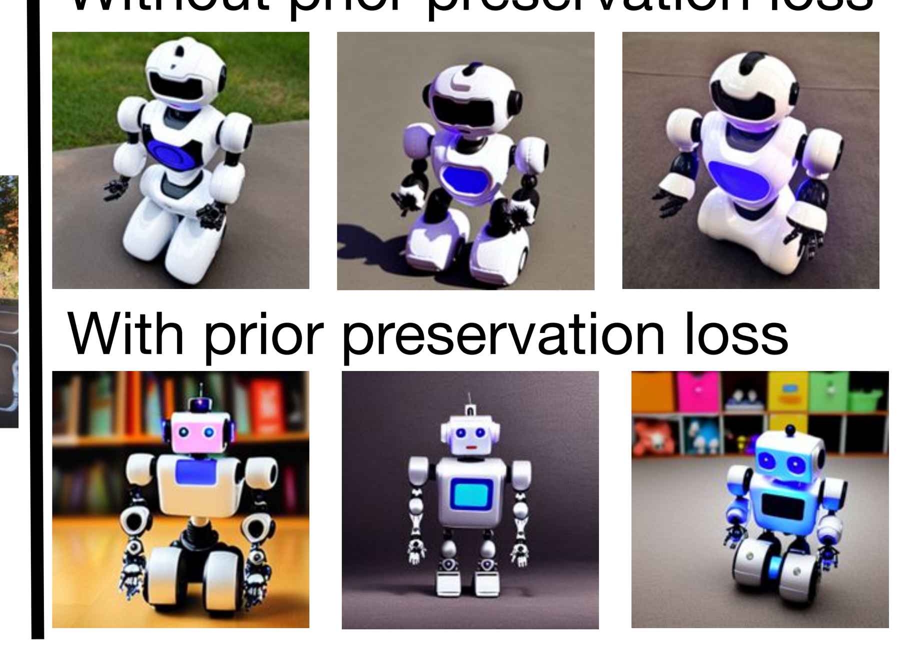
- **Subject fidelity and prompt fidelity.**
  - **Subject fidelity (DINO): similarity** between the **generated** and **real image embeddings.**
  - **Prompt fidelity (CLIP): similarity** between the **prompt** and **generated image** embeddings

Method / Category	DINO	CLIP-I	CLIP-T
Reported paper baselines			
Real Images	0.774	0.885	N/A
DreamBooth (Imagen)	0.696	0.812	0.306
DreamBooth (Stable Diffusion)	0.668	0.803	0.305
Textual Inversion (Stable Diffusion)	0.569	0.780	0.255
Local reproduction results			
bottle	**0.799**	0.817	0.269
cat2	0.768	**0.890**	0.276
dog8	0.666	0.885	0.273
duck_toy	0.714	0.862	**0.326**
robot_toy	0.733	0.842	0.285
Average	0.736	0.859	0.286

Generating a general robot

Input Images

Without prior preservation loss



With prior preservation loss

Without prior preservation loss, diversity is lost

Challenges & Future Work

- Fine-tuned models do not consistently produce good results
- Hyperparameter tuning is tricky as training one pass requires large amount of compute
- Future work: fine-tuning with LoRA to reduce compute and fine-tuning better base models